HUMSAFAR

A Bi-Directional Remote Intervention and Care Coordination System for Vulnerable Individuals

Software Requirements Specification (SRS)

Version 1.0 — Final Locked Baseline

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements of Humsafar, a bi-directional remote intervention and care coordination system for vulnerable individuals. It formalizes the complete, approved feature baseline into a structured specification intended to serve as the foundation for all subsequent project documentation, including the Software Design Specification, System Architecture Document, Database Design, API Specification, and related deliverables. This document is the authoritative reference for the system's required behavior and quality attributes.

1.2 Document Conventions

- **Requirement identifiers** follow a domain-letter convention (e.g., FR-D4) that groups related requirements and supports full traceability to the Requirements Traceability Matrix in Appendix 7.2.
- **Non-Functional Requirement identifiers** follow a sequential NFR-[n] scheme.
- **Priority** is expressed as Core (Must Have), Recommended (Should Have), or Optional (Could Have).
- **"Guardian"** used without qualification refers to either a Primary or Secondary Guardian; privilege distinctions are stated explicitly wherever they differ.
- Requirements are stated using "shall" to denote mandatory system behavior, consistent with IEEE 830 convention.

1.3 Intended Audience and Reading Suggestions

- **Development team:** the complete document, as the primary reference for design and implementation.
- **Project Supervisor and Evaluation Panel:** Sections 1–2 for scope and context; Section 4 for functional detail; Section 5 for measurable quality benchmarks.
- **Future maintainers:** the complete document, as an onboarding reference.

The suggested reading order follows the document's structure. Section 4's domain-grouped subsections (4.1 through 4.7) may be read independently once Sections 1 through 3 have been reviewed.

1.4 Product Scope

Humsafar is a mobile-first care-coordination platform connecting Guardians (Primary and Secondary) with Dependents — vulnerable individuals requiring monitoring and periodic intervention — through layered, largely automated crisis detection, real-time communication, medical record management, and access to external healthcare and home-care services.

In scope: identity and permission management; Guardian–Dependent relationship management; location tracking and geofencing; medication management; layered emergency detection and escalation (missed-dose escalation, Guardian Wellness Check, Panic Mode, remote alarm override); text, voice, and video communication; medical records management; wearable health-band integration; a healthcare service directory; a home-care provider directory with request-to-book functionality; on-demand and event-triggered camera viewing; device health monitoring; prioritized notifications; security, encryption, and data-retention controls; accessibility features; and a caregiver analytics dashboard.

Explicitly out of scope for this project:

- Auto-answer emergency audio link (superseded by Priority Voice Messages)
- Continuous 24/7 CCTV recording and cloud storage (implemented instead as on-demand and event-triggered viewing)
- In-app payments and a formal home-care provider vetting/ratings pipeline
- Voice-guided interaction for the Dependent
- Custom wearable hardware or firmware
- FHIR/HL7 medical-data interoperability
- Formal HIPAA/GDPR-equivalent compliance certification
- Real-time dispatch-system API integration with external healthcare/emergency organizations
- iOS platform support

1.5 References

- Humsafar Project Vision and Problem Statement
- Humsafar Functional and Non-Functional Requirements Specification (initial draft)
- Humsafar Feature Specification and Final Feature Set Report
- Humsafar Feature Set Lock Addendum
- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications

2. Overall Description

2.1 Product Perspective

Humsafar is a new, standalone product; it is not a successor to or extension of an existing in-house system. It operates alongside a set of independent, third-party healthcare and emergency-service organizations (for example, ambulance providers, blood banks, and laboratories), which are referenced and directory-listed within the application but are not technically integrated — no live API access or data-sharing agreement is assumed. The system comprises a Dependent-facing mobile application (including a simplified "Dependent Mode" for safety-critical actions), a Guardian-facing mobile application, a backend server, and an externally sourced wearable health band integrated via Bluetooth Low Energy.

Core Bi-Directional Interaction Model

Guardian → Dependent	Dependent → Guardian
Medication reminders	Live location
Wellness check prompts	Health metrics (manual entry and wearable)
Emergency notifications	Medication confirmations
Communication (messages, calls)	Wellness check responses
Care management (schedules, permissions)	Emergency alerts (Panic Mode)
	Health record updates *

** Health records and metrics may be entered or updated by either the Dependent or the Guardian, subject to the permission dashboard defined in Section 4.1 (FR-A4). This accounts for Dependents who may not reliably self-report.*

Supplementary Interactions

The following are on-demand interactions with the Dependent's environment or with external parties, rather than steady two-way exchanges between Guardian and Dependent, and are treated separately from the core interaction model above:

- **Camera Viewing** — on-demand or event-triggered viewing of the Dependent's environment.
- **Healthcare Directory** — lookup and contact of external healthcare and emergency organizations.
- **Home-Care Marketplace** — browsing and requesting services from directory-listed professionals.

2.2 Product Functions

The following is a high-level summary of the system's major function groups; complete detail is provided in Section 4.

- Identity verification and role-based access control
- Guardian–Dependent linkage and permission management
- Continuous adaptive location tracking and geofencing
- Layered emergency detection and multi-tier escalation

- Medication scheduling, reminders, and adherence tracking
- Text, voice, and video communication
- Medical record storage, export, and appointment tracking
- Wearable-based passive health monitoring
- Healthcare service directory and home-care provider discovery and booking
- On-demand and event-triggered camera viewing
- Device health monitoring (battery and connectivity)
- Prioritized, acknowledgment-tracked notifications
- Encryption, access control, and audit logging
- Accessibility-focused Dependent Mode
- Caregiver-facing analytics dashboard

2.3 User Classes and Characteristics

User Class	Characteristics
Primary Guardian	Registered, identity-verified adult; creates and links Dependent accounts; full permission-granting authority; may monitor multiple Dependents.
Secondary Guardian	Registered, identity-verified adult invited by a Primary Guardian; access limited to modules explicitly granted.
Dependent	Elderly individuals, chronically ill patients, physically disabled individuals, dementia patients, and others requiring continuous care and monitoring; identity-verified; assumed able to operate a smartphone at a basic level, independently or with caregiver assistance; interacts primarily through Dependent Mode for safety-critical actions.
Home-Care Professional	Nurses and physiotherapists listed under the Home-Care Marketplace as directory entries maintained by the platform. Not an authenticated system user; does not hold a platform account in this release.

2.4 Operating Environment

The system targets **Android only**. This reflects Android's stronger Bluetooth Low Energy support (directly relevant to wearable-band integration), more straightforward background-service execution (relevant to continuous location tracking and device monitoring), a simpler overall hardware-integration path, and realistic scope for the project timeline. iOS support is not implemented in this release.

The operating environment consists of:

- A Dependent-facing Android application and a Guardian-facing Android application.
- A backend server providing API access to both applications.
- A Bluetooth Low Energy-compatible wearable health band, integrated via standard BLE GATT profiles.
- An existing, off-the-shelf IP camera for on-demand viewing.

- Active internet connectivity (Wi-Fi or mobile data), required for normal operation; no offline mode is provided.

Platform and SDK requirements: minimum SDK API 26 (Android 8.0 Oreo); target SDK the latest stable Android release available at build time, for Google Play compliance.

2.5 Design and Implementation Constraints

- No offline or degraded-connectivity mode is provided; the system assumes active internet connectivity as its normal operating condition.
- The wearable health band is integrated via standard BLE GATT profiles only; custom wearable hardware or firmware is out of scope.
- Camera functionality assumes an existing, off-the-shelf IP camera; camera hardware is not a deliverable of this project.
- No formal HIPAA/GDPR-equivalent compliance certification is claimed; the system follows general data-protection principles.
- Team size (three developers) and a single academic-year timeline are binding constraints reflected throughout the locked scope.
- The application supports English and Urdu localization.
- The application targets Android only.

2.6 User Documentation

A supplementary Guardian and Dependent user guide may be produced as part of the broader project deliverable set. Its content is outside the scope of this specification.

2.7 Assumptions and Dependencies

- The Dependent's device supports Bluetooth Low Energy, required for wearable band synchronization.
- A compatible wearable band is available to the Dependent, obtained outside the application itself.
- Contact information for external healthcare organizations is publicly available and does not require a formal data-sharing agreement to display.
- Continuous internet connectivity is assumed as the system's normal operating condition.
- Home-Care Professionals listed in the directory are assumed to be vetted through means external to the application.
- Health record and metric entry may be performed by either the Dependent or the Guardian, subject to the permission dashboard.

3. External Interface Requirements

Vendor and technology selections are intentionally left generic throughout this section. This specification defines that an interface is required; the specific product or library selected to fulfill it is a decision for the System Architecture Document.

3.1 User Interfaces

- The system shall provide a native Android graphical user interface, with distinct views for the Guardian role and the Dependent role.
- The Dependent-facing interface shall provide a simplified "Dependent Mode" for three safety-critical actions: Panic Mode, Guardian Wellness Check response, and Medication Confirmation, with minimal navigation depth and large touch targets.
- The interface shall support a large-text, high-contrast display mode, applied at minimum to the three Dependent Mode safety-critical screens.
- The interface shall support English and Urdu localization throughout.
- The Guardian-facing interface shall provide a permission-management dashboard, an interactive map view, a communication interface, a medical-records viewer, and an analytics dashboard.
- Push notifications shall be visually distinguishable by priority tier within the Android notification system.

3.2 Hardware Interfaces

- The system shall interface with a Bluetooth Low Energy-compatible wearable health band via standard BLE GATT profiles — specifically the Heart Rate Service (0x180D) and Battery Service (0x180F). No proprietary or vendor-specific protocol is assumed.
- The system shall interface with the host Android device's GPS/location sensor for continuous location tracking.
- The system shall interface with the host Android device's accelerometer, where the fall-detection feature is implemented.
- The system shall interface with the host Android device's microphone, speaker, and camera for voice/video communication and priority voice messages.
- The system shall interface with an existing, off-the-shelf IP camera supporting a standard live-streaming protocol, for on-demand viewing and event-triggered clip capture.
- The system shall interface with the host device's local audio output at maximum volume for the remote alarm override, subject to Android OS permission and volume-control APIs.

3.3 Software Interfaces

- The system shall interface with a backend server via a defined API.
- The system shall integrate with a third-party identity verification service supporting CNIC validation and liveness detection.
- The system shall integrate with an Android-compatible push notification delivery service.
- The system shall integrate with a real-time communication SDK or service for voice and video calling.
- The system shall integrate with a mapping/geolocation service to render Dependent location on an interactive map.
- The system shall integrate with an SMS and/or voice-call gateway service to deliver the External Emergency Contact Fallback when linked Guardians are unreachable.

- The system shall generate PDF-format documents for medical-record and health-metric export.

3.4 Communications Interfaces

- All communication between the mobile application and the backend server shall occur over an encrypted transport protocol (HTTPS/TLS at minimum).
- Real-time audio/video communication shall use a signaling and media-transport protocol appropriate to the selected real-time communication SDK.
- Push notification delivery shall occur over the selected push service's standard delivery protocol.
- Wearable band communication shall occur over Bluetooth Low Energy, within standard BLE range and pairing constraints.
- The External Emergency Contact Fallback shall communicate over standard SMS and/or PSTN voice-call protocols via the selected gateway service.

4. Functional Requirements

Functional requirements are organized into seven parts by domain. Each requirement is identified by a unique identifier used consistently throughout this document and the Requirements Traceability Matrix (Appendix 7.2).

4.1 Part A — Identity, Access, and Guardian-Dependent Relationship

FR-A1 — Role-Based Registration and Guardian Hierarchy *(Core)*

- The system shall allow a new user to register as either a Guardian or a Dependent.
- The system shall support two Guardian sub-roles: Primary Guardian and Secondary Guardian.
- A Guardian account's role is fixed at first registration and applies uniformly across every Dependent that account is subsequently linked to. An account that registers by independently creating its first Dependent is designated Primary; an account created solely by accepting an invitation is designated Secondary. This designation does not change thereafter.

  Dependencies: Precedes FR-A2/A3 and FR-B1.

  Note: Because the Primary/Secondary designation is fixed at the account level, a Secondary-designated account cannot independently register a new Dependent of its own, and a Primary-designated account cannot accept a Secondary invitation for a different Dependent. This is an accepted design trade-off in favor of a simpler permission model.

FR-A2 — Dependent Identity Verification (KYC) *(Core)*

- The system shall require a Dependent to submit a valid CNIC number during registration.
- The system shall require a Dependent to complete a real-time liveness photo capture during registration.
- The system shall not activate a Dependent account until CNIC validation succeeds, liveness verification succeeds, and the Dependent is linked to exactly one Primary Guardian.

  Dependencies: Gates FR-A1 registration completion; depends on FR-B1 for the linkage condition.

FR-A3 — Guardian Identity Verification *(Core)*

- The system shall verify a Guardian's identity via CNIC validation and/or a verified phone number (OTP) before granting access to the permission dashboard or any intervention capability.
- Guardian verification shall not require liveness photo capture.

  Dependencies: Gates FR-A4 and all Guardian-side intervention features.

FR-A4 — Guardian Permission Dashboard *(Core)*

- The system shall provide the Primary Guardian a dashboard listing every governable module for each linked Dependent.
- The system shall allow the Primary Guardian to independently grant or revoke a Secondary Guardian's access to each module.
- The governable module list shall include, at minimum: Live Location viewing, Health Records, Communication, Emergency Alert Visibility (receiving Panic Mode and Wellness Check escalation notifications), Camera Viewing, and Medication Management. Remote Alarm Override is explicitly excluded from this list, as it is an inherent Primary Guardian privilege that is never delegable to Secondary Guardians under any dashboard configuration.

- Where a Guardian is linked to multiple Dependents, the dashboard shall present permission configuration independently per Dependent.

  Dependencies: Depends on FR-A1/A3; gates access enforcement across nearly all other domains.

FR-B1 — Guardian-Dependent Linkage *(Core)*

- The system shall allow a Guardian account designated Primary to initiate linkage with a new Dependent.
- Guardian accounts designated Secondary shall not be permitted to independently register or initiate linkage with a new Dependent; a Secondary-designated account may only be linked to a Dependent via invitation from that Dependent's Primary Guardian.
- The system shall allow a Primary Guardian to invite additional Guardians to a linked Dependent; invited Guardians shall be designated Secondary for that Dependent.
- A Dependent account shall not be activated until linked to exactly one Primary Guardian.

  Dependencies: Gates FR-A2; feeds FR-B2.

FR-B2 — Guardian Cap Enforcement *(Core)*

- The system shall enforce a maximum of three Guardians per Dependent: exactly one Primary Guardian and up to two Secondary Guardians.
- The system shall reject any linkage attempt that would exceed this cap or introduce a second Primary Guardian for the same Dependent.

  Dependencies: Depends on FR-B1.

FR-B4 — Multi-Dependent Support per Guardian *(Core)*

- The system shall allow a single Guardian account to be linked, as Guardian, to multiple Dependent accounts simultaneously.
- The system shall allow a Guardian to switch between linked Dependents within a unified dashboard view.
- The system shall maintain independent permission configurations per Dependent for a given Guardian.
- A Guardian's Primary/Secondary designation is fixed at the account level and therefore identical across every Dependent linked under this requirement.

  Dependencies: Extends FR-A4.

Permission-change audit logging for this domain is specified under FR-N3 in Section 4.6, where it is consolidated with other sensitive-action logging to avoid duplicate implementation of the same underlying capability.

4.2 Part B — Location, Emergency Detection and Escalation, Device Health

FR-C1 — Continuous Adaptive Location Tracking *(Core)*

- The system shall continuously track the Dependent's device location while the application is active.
- The system shall adjust location update frequency based on the Dependent's movement state — lower frequency while stationary, higher frequency during movement.
- The system shall transmit location updates to the backend server for Guardian access, subject to permission gating (FR-A4).

Dependencies: Feeds FR-C2, FR-D4, FR-L1.

FR-C2 — Geofence Definition and Breach Alerts *(Core)*

- The system shall allow a Primary Guardian to define a geofence for a Dependent. Geofence configuration is Primary-only; Secondary Guardians granted Live Location access may view location and geofence status but not modify zone configuration.
- The system shall generate an immediate high-priority alert to linked Guardians when the Dependent's location falls outside the defined geofence.

Dependencies: Depends on FR-C1.

FR-C3 — Multi-Zone Geofencing *(Recommended)*

- The system shall allow more than one geofence zone to be defined per Dependent.
- The system shall treat the Dependent as within a safe area if their location falls within any defined zone; a breach alert shall be generated only when the location falls outside all defined zones simultaneously.

Dependencies: Extends FR-C2.

FR-C4 — Location History and Route Playback *(Optional)*

- The system shall store a history of the Dependent's location updates.
- The system shall allow a Guardian to view historical location data on a timeline or map interface.

Dependencies: Depends on FR-C1.

FR-D1 — Medication Missed-Dose Escalation *(Core)*

- The system shall detect when a scheduled medication dose is not confirmed within its configured window.
- Upon a missed dose, the system shall send a high-priority alert to linked Guardians.
- The system shall provide the Guardian options to extend the medication time or trigger further escalation.

Dependencies: Depends on FR-F1; feeds FR-D5.

FR-D2 — Panic Mode *(Core)*

- The system shall provide the Dependent a single, easily accessible control within Dependent Mode to trigger an emergency alert.
- Activating Panic Mode shall instantly notify all linked Guardians, regardless of individual module permissions — a deliberate safety-critical exception to standard permission gating.
- Panic Mode is a supplementary detection layer and is not the system's primary or sole crisis-detection mechanism.

Dependencies: Independent trigger; no upstream dependency.

FR-D3 — Remote Alarm Override *(Core)*

- The system shall allow a Primary Guardian to remotely trigger a high-decibel alarm on the Dependent's device.
- The alarm shall override the Dependent device's local volume/silent state, subject to Android OS permission APIs.
- Remote Alarm Override is a Primary-Guardian-exclusive privilege and is not delegable to Secondary Guardians under any permission-dashboard configuration.

Dependencies: No upstream dependency.

FR-D4 — Guardian Wellness Check *(Core)*

- The system shall allow a Guardian to configure one or more scheduled wellness-check windows for a Dependent (recommended default: up to three per day).
- At each scheduled time, the system shall prompt the Dependent, via Dependent Mode, to confirm their well-being.
- If unconfirmed within a configured grace period, the system shall check the Dependent's recent device activity before escalating.
- If no recent activity is detected, the system shall escalate per the multi-level escalation chain (FR-D5).
- The system shall provide a default wellness-check schedule, editable by the Guardian.

Dependencies: Depends on FR-C1, FR-L1; feeds FR-D5.

FR-D5 — Multi-Level Escalation Chain *(Recommended)*

- The system shall escalate an unacknowledged critical alert from the Primary Guardian to Secondary Guardians if unacknowledged within a configured timeout.
- If no linked Guardian acknowledges the escalated alert within a further configured timeout, the system shall notify the External Emergency Contact.

Dependencies: Depends on FR-D1, FR-D4, and FR-M2; feeds FR-R1.

FR-D6 — Fall Detection *(Optional)*

- The system shall analyze accelerometer data from the Dependent's device, using a threshold-based model, to detect a probable fall event.
- Upon a probable fall detection, the system shall notify linked Guardians with a "possible fall" alert.

Note: Detection is probabilistic, not guaranteed; accuracy claims should be scoped conservatively in project documentation.

FR-R1 — External Emergency Contact Fallback *(Recommended)*

- The system shall allow a Primary Guardian to designate one external emergency contact for a Dependent, outside the application's own Guardian accounts.
- If the escalation chain exhausts all linked Guardians without acknowledgment, the system shall notify the external emergency contact via SMS and/or voice call.

Dependencies: Depends on FR-D5 and the SMS/voice gateway interface (Section 3.3).

FR-L1 — Battery Monitoring, Low-Battery Alert, and Last-Known Location *(Core)*

- The system shall continuously monitor the Dependent device's battery level.
- The system shall notify linked Guardians when battery falls to 15%.

- The system shall capture and store the Dependent's last-known location immediately prior to anticipated device shutdown.

  Dependencies: Feeds FR-D4; depends on FR-C1.

FR-L2 — Connectivity-Loss Alert *(Recommended)*

- The system shall detect when a Dependent's device has been offline beyond a configured threshold duration.
- The system shall notify linked Guardians of prolonged connectivity loss, distinct from the low-battery alert.

4.3 Part C — Communication and Notifications

FR-E1 — Text Messaging *(Core)*

- The system shall allow a Guardian and a linked Dependent to exchange text messages.

  Dependencies: Depends on FR-B1.

FR-E2 — Voice and Video Calling *(Core)*

- The system shall allow a Guardian and a linked Dependent to initiate real-time voice calls.
- The system shall allow a Guardian and a linked Dependent to initiate real-time video calls.
- Calls shall require the receiving party to actively accept before audio or video is established.

  Dependencies: Depends on FR-B1; interfaces per Section 3.3.

FR-E4 — Priority Voice Messages *(Core)*

- The system shall allow a Guardian to record and send a voice message flagged as high priority.
- Priority voice messages shall be delivered as high-priority notifications, distinct from standard messages.
- Priority Voice Messages are the system's sole asynchronous urgent-audio mechanism.

  Dependencies: Depends on FR-B1; feeds FR-M1.

  Note: This mechanism is one-directional and does not provide live two-way audio without the Dependent actively answering a call. This is an accepted, documented limitation.

FR-E5 — Multi-Guardian Bridge Call *(Optional)*

- The system shall allow all Guardians linked to a Dependent to be joined into a single call during an active emergency.

  Dependencies: Depends on FR-E2, FR-D2, and FR-D5 for emergency-state definition.

FR-M1 — Priority-Tiered Push Notifications *(Core)*

- The system shall deliver push notifications to Guardians and Dependents via the Android notification system.
- The system shall support at least two priority tiers — standard and critical — visually distinguishable.
- Critical-tier notifications shall be used for: Panic Mode, Wellness Check escalation, geofence breach, missed-medication escalation, low battery, connectivity loss, and Priority Voice Messages.

  Dependencies: Feeds and is depended on by nearly every other domain.

FR-M2 — Notification Acknowledgment Tracking *(Recommended)*

- The system shall record whether a critical-tier notification has been opened/acknowledged by its recipient, not merely delivered.
- Unacknowledged critical notifications shall be surfaced to the escalation logic (FR-D5) as a "no response" state.

  Dependencies: Feeds FR-D5.

4.4 Part D — Medication, Medical Records, and Health Metrics

FR-F1 — Medication Scheduling and Dual Confirmation *(Core)*

- The system shall allow a Guardian to define a medication schedule for a Dependent (drug name, dosage, time(s), frequency).
- The system shall send a reminder notification to the Dependent before each scheduled dose time.
- The system shall require a two-step confirmation from the Dependent to mark a dose as taken.
- The system shall send a high-priority alert as the dose time approaches if not yet confirmed.
- If a dose remains unconfirmed past its configured window, the system shall trigger FR-D1.

Dependencies: Feeds FR-D1, FR-Q1.

FR-G1 — Medical Vault and PDF Export *(Core)*

- The system shall store prescriptions, laboratory reports, X-rays, and other medical records for a Dependent.
- Records may be entered or uploaded by either the Dependent or the Guardian, subject to permission gating.
- The system shall generate PDF exports of medical records in three forms: complete history, year-wise, and month-wise.

Dependencies: Depends on FR-N2; feeds FR-G3, FR-Q1, FR-Q2.

FR-G2 — Field-Level Access Control on Records *(Recommended)*

- The system shall allow the Primary Guardian to restrict, per record category, which categories a Secondary Guardian may view.

Dependencies: Extends FR-A4; depends on FR-G1.

FR-G3 — Doctor-Shareable Read-Only Export Link *(Optional)*

- The system shall allow a Guardian or Dependent to generate a read-only shareable link to a Dependent's medical records.
- The link shall expire after a configurable period (recommended default: 72 hours).

Dependencies: Depends on FR-G1.

FR-G4 — Appointment and Event Tracker *(Core)*

- The system shall allow scheduling of doctor appointments for a Dependent.
- The system shall generate reminder notifications for scheduled appointments to both Guardian and Dependent.

Dependencies: Feeds FR-M1.

FR-H1 — Manual Daily Vitals Entry *(Core)*

- The system shall allow manual recording of blood pressure, blood sugar, oxygen saturation, and body temperature for a Dependent.
- Vitals may be entered by either the Dependent or the Guardian.
- Health-metric reports shall be exportable as a separate PDF, or combined with the complete medical history export.

Dependencies: Feeds FR-Q1; depends on FR-N2.

FR-H2 — Wearable Health Band Integration *(Core (Mandated))*

- The system shall integrate with a Bluetooth Low Energy wearable health band via standard GATT profiles (Heart Rate Service 0x180D, Battery Service 0x180F).
- The system shall periodically sync heart-rate and activity/step-count data from the paired wearable band.
- Wearable-derived data shall feed FR-Q1 and may optionally inform FR-D4 as a supplementary signal; it shall not serve as a sole crisis-detection trigger.
- The system shall handle wearable disconnection or out-of-range conditions gracefully, without blocking other application functionality.

Dependencies: Depends on Section 3.2; feeds FR-Q1, optionally FR-D4.

4.5 Part E — Healthcare Directory and Home-Care Marketplace

FR-I1 — Healthcare Service Directory *(Core)*

- The system shall provide a directory of healthcare and emergency service organizations with contact information and service details.
- The system shall support nearby-organization discovery based on the Dependent's or Guardian's location.
- The system shall provide quick-access contact initiation for listed organizations from both the Dependent Mode and Guardian interfaces.

  Dependencies: Directory content maintained by the platform; no live third-party dispatch-API integration is assumed.

FR-J1 — Home-Care Provider Directory and Request-to-Book *(Core)*

- The system shall provide a directory of home-care professionals with profile, specialization, and service-charge information.
- Home-Care Professionals shall be directory-listed entries maintained by the platform; they shall not hold platform accounts or authenticate directly.
- The system shall allow a Guardian or Dependent to send a booking request to a listed provider.

  Dependencies: Feeds the shared Activity Log (FR-I2/J2).

FR-I2/J2 — Shared Activity Log *(Recommended)*

- The system shall log each instance of a healthcare organization being contacted or a home-care booking request being made, recording the initiating party, timestamp, and organization/provider.
- The system shall present this log as a single, unified Activity Timeline to linked Guardians.

  Dependencies: Depends on FR-I1 and FR-J1.

4.6 Part F — Surveillance, Security-Related Functions, and Data Management

FR-K1 — On-Demand Live Camera View *(Core)*

- The system shall allow an authorized Guardian to request an on-demand live video stream from a Dependent's paired IP camera.
- Camera access shall be gated by Guardian-dashboard permission.
- The Dependent's device shall display a visible on-screen indicator whenever the camera is being actively viewed by a Guardian, providing notice that monitoring is in progress.

Dependencies: Depends on Section 3.2.

FR-K2 — Event-Triggered Clip Capture *(Recommended)*

- The system shall automatically capture and store a short video clip from the paired camera when a critical event occurs (Panic Mode or Remote Alarm activation).
- Captured clips shall be retained per the data retention policy (FR-O1).

Dependencies: Depends on FR-K1, FR-D2, FR-D3, FR-O1.

FR-N1 — End-to-End Encryption, Audio/Video *(Core)*

- The system shall encrypt all real-time audio and video streams end-to-end.

Dependencies: Depends on FR-E2, FR-K1.

FR-N2 — Encryption at Rest *(Core)*

- The system shall encrypt stored medical records, health metrics, and location history at rest.

Dependencies: Applies to FR-G1, FR-H1, FR-H2, FR-C1, FR-C4.

FR-N3 — Sensitive-Action Audit Logging *(Recommended)*

- The system shall log all permission changes, camera-view events, and medical-record access events, recording actor, action, target, and timestamp.
- Audit logs shall be viewable by the Primary Guardian for their linked Dependents.

Dependencies: Consolidates the permission-change audit requirement from Section 4.1; depends on FR-A4, FR-K1, FR-G1.

FR-O1 — Data Retention and Deletion Policy *(Recommended)*

- The system shall retain Dependent data for the duration the account remains active.
- Upon account closure, the system shall delete or anonymize retained data within a configurable period (recommended default: 60 days), except where retention is required for an active audit or dispute.

Dependencies: Applies across FR-G1, FR-K2, FR-N3, FR-C4.

4.7 Part G — Accessibility and Analytics

FR-P1 — Dependent Mode (Simplified UI) *(Core)*

- The system shall provide a simplified navigation mode for the Dependent, scoped at minimum to Panic Mode, Wellness Check response, and Medication Confirmation.
- Dependent Mode screens shall require minimal navigation depth and use large touch targets.

  Dependencies: Applies to FR-D2, FR-D4, FR-F1.

FR-P2 — Large Text and High-Contrast Mode *(Core)*

- The system shall provide a large-text, high-contrast display mode, applied at minimum to FR-P1's three safety-critical screens.

  Dependencies: Extends FR-P1.

FR-P4 — Multilingual (Urdu) Support *(Core)*

- The system shall support English and Urdu localization for all user-facing text.

FR-Q1 — Adherence and Vitals Analytics Dashboard *(Recommended)*

- The system shall present medication adherence trends to the Guardian.
- The system shall present vitals trends, derived from manual and wearable data, to the Guardian.

  Dependencies: Depends on FR-F1, FR-H1, FR-H2.

FR-Q2 — Automated Summary Reports *(Optional)*

- The system shall generate a scheduled (weekly or monthly) summary report for the Guardian, combining adherence and vitals trends with the PDF export mechanism.

  Dependencies: Depends on FR-Q1, FR-G1.

Voice-Guided Interaction, present in earlier project iterations, is out of scope for this release and is not specified further in this document; it is retained in the Requirements Traceability Matrix (Appendix 7.2) for completeness.

5. Non-Functional Requirements

NFR-01 — Command Latency (Performance). Remote command executions (Remote Alarm Override, Panic Mode notification) shall initiate on the Dependent's device within 2 seconds of the Guardian's command, over a standard 4G network connection.

NFR-02 — Background Telemetry Battery Efficiency (Performance). Background location polling and low-priority telemetry monitoring shall not consume more than 5% of the Dependent device's battery per hour. This budget applies to background/idle-state telemetry only; active sessions (video calls, live camera viewing, active Wellness Check/Panic Mode interaction) are excluded, given their inherently higher resource cost.

NFR-03 — Communication Confidentiality (Security). All real-time audio and video streams shall be encrypted end-to-end.

NFR-04 — Data-at-Rest Confidentiality (Security). All stored medical records, health metrics, and location history shall be encrypted at rest.

NFR-05 — System Availability (Reliability). The backend server shall maintain 99.9% uptime, measured monthly (permitting approximately 43 minutes of downtime per month). This is not a zero-downtime guarantee. During backend unavailability, the mobile application shall queue outgoing critical alerts locally and transmit them immediately upon reconnection.

NFR-06 — Scalability. The system shall support exactly 3 concurrent Guardian connections monitoring a single Dependent without performance degradation.

NFR-07 — Usability, Dependent Mode. Each of Dependent Mode's three safety-critical actions shall be completable within 2 taps from the Dependent Mode home screen.

NFR-08 — Accessibility Standard (Usability). The Large Text/High-Contrast mode shall meet a minimum contrast ratio consistent with WCAG 2.1 Level AA guidance on the three safety-critical screens, referenced as a best-practice benchmark rather than a formal compliance certification.

NFR-09 — Localization Completeness (Usability). 100% of user-facing static text shall be available in both English and Urdu; no screen shall present untranslated text in either supported language.

NFR-10 — Data Privacy Principles (Compliance). The system shall follow general data-protection principles — data minimization, purpose limitation, access control — in handling Dependent data. No formal HIPAA/GDPR-equivalent certification is claimed.

NFR-11 — Maintainability. The codebase shall be organized so each domain-grouped module (Section 4, Parts A through G) can be developed and tested independently.

NFR-12 — Portability. The application shall run on Android devices from API 26 (Android 8.0) through the latest stable Android release available at build time. iOS portability is out of scope.

NFR-13 — Wearable Sync Reliability (Reliability). The system shall attempt automatic reconnection to a previously paired wearable band within 30 seconds of it returning to range, without requiring manual user intervention.

6. System Models

6.1 Actors

Actor	Description
Primary Guardian	Registered, verified Guardian who created a Dependent's account; full permission authority; exclusive access to Remote Alarm Override.
Secondary Guardian	Registered, verified Guardian invited by a Primary Guardian; access scoped by dashboard permissions.
Dependent	Registered, KYC-verified vulnerable individual; primary interaction via Dependent Mode.
External Emergency Contact	Non-account-holding fallback recipient of last-resort SMS/call notifications. Not an authenticated system user.
Home-Care Professional	Directory-listed entity; does not authenticate or interact with the system directly.
Backend System	The automated actor responsible for escalation timing, notification delivery, and scheduled triggers.

6.2 Use Case Summary

ID	Use Case	Primary Actor(s)	Related FR(s)
UC-01	Register as Guardian or Dependent	Primary/Secondary Guardian, Dependent	FR-A1, FR-A2, FR-A3
UC-02	Link Dependent to Primary Guardian	Primary Guardian, Dependent	FR-B1, FR-B2
UC-03	Invite Secondary Guardian	Primary Guardian	FR-B1
UC-04	Configure permission dashboard	Primary Guardian	FR-A4
UC-05	View live location	Guardian	FR-C1, FR-C2
UC-06	Configure geofence(s)	Primary Guardian	FR-C2, FR-C3
UC-07	Configure medication schedule	Guardian	FR-F1
UC-08	Confirm medication taken	Dependent	FR-F1
UC-09	Configure Wellness Check schedule	Guardian	FR-D4
UC-10	Respond to Wellness Check	Dependent	FR-D4

UC-11	Trigger Panic Mode	Dependent	FR-D2
UC-12	Trigger Remote Alarm Override	Primary Guardian	FR-D3
UC-13	View live camera feed	Guardian	FR-K1, FR-K2
UC-14	Communicate (message/call)	Guardian, Dependent	FR-E1, FR-E2, FR-E4
UC-15	Manage medical records	Guardian, Dependent	FR-G1, FR-G2, FR-G3, FR-G4
UC-16	Record daily vitals	Guardian, Dependent	FR-H1
UC-17	Sync wearable data	Dependent, Backend System	FR-H2
UC-18	Browse healthcare directory	Guardian, Dependent	FR-I1
UC-19	Request home-care service	Guardian, Dependent	FR-J1
UC-20	View analytics dashboard	Guardian	FR-Q1, FR-Q2
UC-21	Receive emergency escalation	Secondary Guardian, External Emergency Contact	FR-D5, FR-R1
UC-22	View audit log	Primary Guardian	FR-N3

6.3 Actor Relationships (Core Safety Subset)

The following summarizes the primary actor-to-use-case relationships for the system's core safety loop. A complete UML use-case diagram is produced as part of the Software Design Specification; the table in 6.2 above is authoritative for full use-case coverage.

Primary Guardian: Link Dependent (UC-02), Configure Permissions (UC-04), Configure Geofence (UC-06), Trigger Remote Alarm (UC-12), View Camera Feed (UC-13)

Secondary Guardian: View Live Location (UC-05), Receive Escalation (UC-21)

Dependent: Confirm Medication (UC-08), Respond to Wellness Check (UC-10), Trigger Panic Mode (UC-11)

Backend System: Sync Wearable Data (UC-17), Drive Escalation Delivery (UC-21)

External Emergency Contact: Receive Escalation (UC-21)

7. Appendices

7.1 Glossary

Term	Definition
Guardian	A registered caregiver account; either Primary or Secondary.
Primary Guardian	The Guardian who created a Dependent's account; holds full permission authority for that Dependent and exclusive access to Remote Alarm Override.
Secondary Guardian	A Guardian invited by a Primary Guardian; access scoped per module via the permission dashboard.
Dependent	A vulnerable individual monitored and supported through the system.
Dependent Mode	The simplified, large-target navigation mode presented to the Dependent for safety-critical actions.
KYC	Know Your Customer — identity verification via CNIC and, for Dependents, liveness photo capture.
Guardian Wellness Check	Scheduled, Guardian-configured prompts requiring Dependent confirmation of well-being, escalating on non-response.
Panic Mode	Dependent-initiated emergency alert control; a supplementary, not primary, crisis-detection layer.
Escalation Chain	The multi-tier notification sequence triggered by an unacknowledged critical alert.
BLE	Bluetooth Low Energy — the wireless protocol used for wearable band integration.
GATT	Generic Attribute Profile — the standard BLE data-exchange format used to read heart-rate and battery data.
Geofence	A Guardian-defined geographic boundary that triggers an alert if the Dependent's location falls outside it.
Activity Log	The shared, unified interaction log for Healthcare Directory contacts and Home-Care Marketplace booking requests.
Out of Scope	Explicitly deferred from this release: the auto-answer emergency audio link, in-app payments and provider vetting, continuous CCTV recording, voice-guided interaction, custom wearable hardware, FHIR/HL7 interoperability, and formal HIPAA/GDPR certification.

7.2 Requirements Traceability Matrix

Maps each approved feature to its Functional Requirement(s), Use Case(s), and governing Non-Functional Requirement(s).

Feature	Priority	FR ID(s)	Use Case(s)	NFR(s)
Role-Based Registration and Guardian Hierarchy	Core	FR-A1	UC-01	NFR-12
Dependent KYC	Core	FR-A2	UC-01	NFR-04
Guardian Identity Verification	Core	FR-A3	UC-01	—
Guardian Permission Dashboard	Core	FR-A4	UC-04	—
Guardian-Dependent Linkage	Core	FR-B1	UC-02, UC-03	—
Guardian Cap Enforcement	Core	FR-B2	UC-02	NFR-06
Permission-Change Audit Log	Recommended	FR-N3	UC-22	—
Multi-Dependent per Guardian	Core	FR-B4	UC-02, UC-04	—
Continuous Adaptive Location Tracking	Core	FR-C1	UC-05	NFR-02
Geofence Definition and Breach Alerts	Core	FR-C2	UC-06	NFR-02
Multi-Zone Geofencing	Recommended	FR-C3	UC-06	—
Location History / Route Playback	Optional	FR-C4	UC-05	NFR-04
Medication Missed-Dose Escalation	Core	FR-D1	UC-08	NFR-01
Panic Mode	Core	FR-D2	UC-11	NFR-01, NFR-07
Remote Alarm Override	Core	FR-D3	UC-12	NFR-01
Guardian Wellness Check	Core	FR-D4	UC-09, UC-10	NFR-01, NFR-07
Multi-Level Escalation Chain	Recommended	FR-D5	UC-21	NFR-01

Fall Detection	Optional	FR-D6	UC-11	—
Text Messaging	Core	FR-E1	UC-14	—
Voice and Video Calling	Core	FR-E2	UC-14	NFR-03
Auto-Answer Emergency Audio Link	Removed	—	—	—
Priority Voice Messages	Core	FR-E4	UC-14	—
Multi-Guardian Bridge Call	Optional	FR-E5	UC-14	—
Medication Scheduling and Dual Confirmation	Core	FR-F1	UC-07, UC-08	—
Adherence History and Reporting	Recommended	FR-Q1 (absorbed)	UC-20	—
Medical Vault and PDF Export	Core	FR-G1	UC-15	NFR-04
Field-Level Access Control on Records	Recommended	FR-G2	UC-15	—
Doctor-Shareable Read-Only Export Link	Optional	FR-G3	UC-15	—
Appointment and Event Tracker	Core	FR-G4	UC-15	—
Manual Daily Vitals Entry	Core	FR-H1	UC-16	NFR-04
Wearable Health Band Integration	Core (Mandated)	FR-H2	UC-17	NFR-13
Healthcare Service Directory	Core	FR-I1	UC-18	—
Contact-Logging / Incident Timeline	Recommended	FR-I2/J2	UC-18, UC-19	—
Home-Care Provider Directory and Request-to-Book	Core	FR-J1	UC-19	—
Session History Log	Recommended	FR-I2/J2	UC-19	—
In-App Payments and Formal Vetting	Future Scope	—	—	—
On-Demand Live	Core	FR-K1	UC-13	NFR-03

Camera View				
Event-Triggered Clip Capture	Recommended	FR-K2	UC-13	—
Continuous 24/7 CCTV Recording	Future Scope	—	—	—
Battery Monitoring and Low-Battery Alert	Core	FR-L1	UC-05	—
Connectivity-Loss Alert	Recommended	FR-L2	UC-05	—
Priority-Tiered Push Notifications	Core	FR-M1	cross-cutting	NFR-01
Notification Acknowledgment Tracking	Recommended	FR-M2	UC-21	—
End-to-End Encryption (Audio/Video)	Core	FR-N1	UC-13, UC-14	NFR-03
Encryption at Rest	Core	FR-N2	UC-15, UC-16, UC-05	NFR-04
Sensitive-Action Audit Logging	Recommended	FR-N3	UC-22	—
Data Retention and Deletion Policy	Recommended	FR-O1	cross-cutting	NFR-10
Dependent Mode (Simplified UI)	Core	FR-P1	UC-08, UC-10, UC-11	NFR-07
Large Text and High-Contrast Mode	Core	FR-P2	UC-08, UC-10, UC-11	NFR-08
Voice-Guided Interaction	Future Scope	FR-P3 (listed only)	—	—
Multilingual (Urdu) Support	Core	FR-P4	cross-cutting	NFR-09
Adherence and Vitals Analytics Dashboard	Recommended	FR-Q1	UC-20	—
Automated Summary Reports	Optional	FR-Q2	UC-20	—
External Emergency Contact Fallback	Recommended	FR-R1	UC-21	—